RUBÉN BARONA ALBA

Frontend Engineer | Angular · TypeScript · JavaScript | Machine Learning
Córdoba, Spain | barona.ruben@gmail.com | +34 665 90 39 88 | linkedin.com/in/ruben-barona-alba | rubenbarona.es

PROFESSIONAL SUMMARY

Frontend engineer specialized in Angular and TypeScript, taking features from specification to production on a public transport platform for the Andalusian Regional Government (Junta de Andalucía). Experienced with Angular, TypeScript, RxJS and Jest, and with React Native, Expo and Firebase on mobile. I focus on maintainable code, efficient reactive data flows and automated testing, reaching 90% coverage with Jest. My Big Data background lets me turn machine learning models into products: web dashboards and interactive apps built on simulation, clustering and similarity models. I use AI agents (Claude Code, Codex) and MCP to speed up scaffolding, test generation and refactoring, while keeping ownership of validation, code quality and technical decisions.

TECHNICAL SKILLS

Frontend: Angular, TypeScript, JavaScript (ES6+), RxJS, HTML5, CSS3, SCSS, Angular Material, Reactive Forms, Responsive Design, in-browser data visualization, DevTools
Mobile: React Native, Expo, Firebase
Testing and quality: Jest (unit and integration tests), SonarQube, code review, bug fixing with QA
APIs and CI/CD: REST APIs, Swagger/OpenAPI, GitLab, Jenkins, Git, Scrum, Trello
AI and machine learning: Random Forest, feature selection (Boruta), clustering, dimensionality reduction, cosine similarity, Poisson models, Monte Carlo, AI agent development (Claude Code, Codex), MCP, n8n
Data: Python (pandas, NumPy, scikit-learn), SQL Server, PostgreSQL, SQLite, InfluxDB, web scraping (Playwright), Streamlit, Grafana, Power BI

PROFESSIONAL EXPERIENCE

Frontend Engineer | ICCA · Córdoba, Spain — Nov 2024 - Oct 2026

- Built end-to-end components and services in Angular and TypeScript for CGCTPA, the public transport management platform of the Andalusian Regional Government, and its SIU real-time passenger information module.
- Integrated every feature with the backend REST API, validating the endpoints in Swagger before implementing them in the service layer.
- Raised test coverage to 90% with Jest (unit and integration tests), reducing regressions in production.
- Cut response times of key components by 20% by optimizing observable streams with RxJS.
- Implemented reactive forms with async validation (debounceTime, switchMap) and integrated Angular Material and the corporate component library.
- Worked in Scrum sprints with team estimation and detailed functional specifications, fixing QA-reported bugs, with a CI/CD pipeline on GitLab, Jenkins and SonarQube.

Junior Developer | Deuser (Indra Group) · Córdoba, Spain — Oct 2023 - Jul 2024

- Built a real-time energy monitoring solution with InfluxDB and Grafana that cut incident detection time from 10 minutes to 1.
- Automated supply chain processes on a SQL Server database with over 500,000 records (Siemens Opcenter APS) using SQL stored procedures and Python and C# scripts.
- Integrated real-time sensor data into SCADA (WinCC OA) for 46 AENA airports.

AI, DATA AND PRODUCT PROJECTS

Season forecasting dashboard, Córdoba CF | Personal project · JavaScript, Python

- Built a dependency-free JavaScript web dashboard that runs a Poisson model with Bayesian regularization and 100,000 Monte Carlo simulations in the browser to project promotion, playoff and relegation odds.
- Automated matchday-by-matchday updates with a Python pipeline (Playwright, pandas) that feeds the dashboard.

Machine learning squad planning, Málaga CF 2026/27 | Personal project — 2026

- Built a Python pipeline that scrapes 9,725 players from 18 competitions (119 metrics), with concurrent requests, resumable checkpoints and offline automated tests.
- Trained position-specific Random Forest models with Boruta feature selection to learn the club profile and ranked candidates by cosine similarity: 8 signings with an average fit score of 90.4%.

Smart scouting system, RC Celta | Master's Thesis — 2024 - 2025

- Designed (team project) a supervised and unsupervised machine learning system on top of the club data provider's API: automatic selection of KPIs from over 1,300 variables, archetype clustering and multivariate similarity.
- Deployed it as a web app in Streamlit with login, filters, radar charts, tactical fit ranking and automated PDF reports.

Freelance web development | Online stores and landing pages · WordPress, WooCommerce, JavaScript

EDUCATION AND LANGUAGES

Master's in Big Data applied to Football Scouting | UCAM · Sports Data Campus — 2024 - 2025
BSc in Computer Engineering, Software Engineering | University of Córdoba (UCO) — 2019 - 2024
Erasmus+, Computer Engineering | Politechnika Opolska · Poland — 2022 - 2023
Languages: Spanish (native) | English B2